

Assignment nr. 3 – Radiation

Assigned: Thursday April 2nd, 2026
Due: Tuesday April 21th, 2026 at class starting

Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

- to solve **individually** the exercises;
- to submit the solution **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal

Problem nr. 1

Let consider an ideal “pencil-beam” antenna whose normalized radiation pattern can be approximated as

$$f(\theta, \varphi) = \begin{cases} 1 & \text{for } \theta < \theta_0 \\ 0 & \text{for } \theta > \theta_0 \end{cases}$$

Find the relation between the maximum gain G and the (half) beam-width θ_0 considering energetic efficiency $\eta = 1$ and plot the behavior of G , as a function of θ_0 , assuming that θ_0 varies in the range $[1^\circ, 90^\circ]$.

Comment the results.

Problem nr. 2

The field radiated by a patch antenna laying in the (y,z) plane can be approximated as follows:

$$\underline{e}(\theta, \phi) = -jV_0 8\pi \frac{W}{\lambda_0} \hat{\phi} \sin \theta \operatorname{sinc}\left(\pi \frac{W}{\lambda_0} \cos \theta\right) \cos\left(\pi \frac{L}{\lambda_0} \sin \theta \sin \phi\right)$$

Your task is to plot - in cartesian format,
- in polar format

- the normalized radiation pattern $e_{norm}(\theta, \phi) = \frac{|\underline{e}(\theta, \phi)|}{e_{max}}$, being $e_{max} = |\underline{e}(\pi/2, 0)|$
- $e_{norm}(\theta, \phi)|_{dB}$

in the two half planes:

- (x,z) , with $x \geq 0$
- (x,y) , with $x \geq 0$

with the following values for the length and the width of the patch L, W : $L/\lambda_0 = 0.23$ and $W/\lambda_0 = 0.52$

Determine the E and H planes.

WARNING! If Matlab will be used to plot the radiation pattern, pay attention to the definition of the sinc(x) implemented:

$$\text{sinc}(x) = \begin{cases} \frac{\sin \pi x}{\pi x} & \text{for } x \neq 0 \\ 1 & \text{for } x = 0 \end{cases}$$

in the present case the argument of the sinc() is therefore $\frac{W}{\lambda_0} \cos \theta$.

Problem nr. 3

The field radiated by a dipole with length $\lambda_0/2$ at the design frequency f_0 , and directed along z can be approximated by the expression

$$\underline{e}(\theta, \phi) = 2jZ_0 I_0 (\sin \theta)^{3/2} \hat{\theta}$$

where I_0 is the current at the input port of the dipole.

- Compute the radiated power as a function of the input current I_0 ;
- Compute the radiation resistance R_{rad}^{dip} of the dipole.

Note: recall that $\int \sin^4 x dx = \frac{3}{8}x - \frac{\sin 2x}{4} + \frac{\sin 4x}{32}$

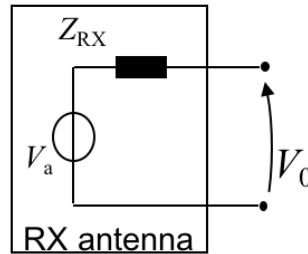
Problem nr. 4

A parabolic antenna has maximum gain $G = 38$ dB and input power $P_{in} = 12$ W.

Compute the magnitude of the electric field $|\underline{E}|$ in the direction of maximum radiation at a distance of 10 km.

Problem nr. 5

Let consider an antenna in the receiving mode. In the direction of arrival of the incident field, $\underline{E}_{inc}$ and the receiving antenna are equi polarized and the gain G of the antenna is maximum.



Compute the amplitude of the open circuit voltage V_0 at the port of the receiving antenna.

Data:

$$f_0 = 10 \text{ GHz} \quad G = 25 \text{ dB} \quad |\underline{E}_{inc}| = 2 \text{ mV/m} \quad Z_{RX} = (75 - j25)\Omega$$